

Disproof of Conjecture 00000000603: the Conway polynomial of $T(2, 25)$ does not peak at $\lfloor (p-1)(q-1)/4 \rfloor$

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Abstract

Conjecture 00000000603 asserts that for every torus knot $T(p, q)$ the coefficient sequence of the Conway polynomial $\nabla(z)$ is nonnegative, has no internal zeros, is log-unimodal, and has its peak at position $\lfloor (p-1)(q-1)/4 \rfloor$. We show that the peak-position clause is false. For $T(2, 25)$ the coefficients of $z^0, z^2, \dots, z^{24}$ are 1, 78, 1001, 5005, 12870, 19448, 18564, $\dots$, so the peak is at index 5 (the term z^{10}), while $\lfloor 1 \cdot 24/4 \rfloor = 6$. For $T(2, 61)$ the predicted position 15 fails under every reasonable indexing convention. Everything is verified in a self-contained Lean 4 project and an independent Python script.

1 The conjecture

The statement (English version) reads:

Definition: The specialization $P_K(1, z)$ of the HOMFLY polynomial $P_K(a, z)$ is the Conway polynomial $\nabla_K(z)$. *Conjecture:* For the torus knot $T(p, q)$, the coefficient sequence of $\nabla(z)$ is nonnegative, has no internal zeros, and is unimodal on the logarithmic scale; the peak position is $\lfloor (p-1)(q-1)/4 \rfloor$.

The Chinese version says the same thing. The statement is a conjunction about every torus knot, so a single torus knot where the peak is not at $\lfloor (p-1)(q-1)/4 \rfloor$ refutes it.

Readings of “peak position”. The Conway polynomial of a knot contains only even powers of z , and $\deg \nabla_{T(p,q)} = 2g = (p-1)(q-1)$. So $\lfloor (p-1)(q-1)/4 \rfloor$ is naturally an index into the sequence of even coefficients. We consider three readings of “the coefficient at position i ”:

- (R1) the coefficient of z^i ;
- (R2) the coefficient of z^{2i} (even coefficients, indexed from 0);
- (R3) the coefficient of $z^{2(i-1)}$ (even coefficients, indexed from 1).

Reading (R2) is the natural one: it is the only one under which the formula holds for all of $T(2, 3), \dots, T(2, 19)$ (Table 1). We use the weakest meaning of “the peak is at i ”: the coefficient at position i is a maximum, with ties allowed. We show that a strictly larger coefficient exists.

2 The Conway polynomial of $T(2, n)$

Let L_n be the closure of the 2-strand braid σ_1^n . For odd n , L_n is the torus knot $T(2, n)$. Also L_0 is the two-component unlink and L_1 is the unknot. The Conway polynomial satisfies $\nabla(\text{unknot}) = 1$, $\nabla(\text{split link}) = 0$, and the skein relation $\nabla(L_+) - \nabla(L_-) = z \nabla(L_0)$. Apply the skein relation at one crossing of σ_1^{n+2} . Changing that crossing gives σ_1^n , and smoothing it gives σ_1^{n+1} . Hence

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = F_n + z F_{n+1}, \quad F_n := \nabla(L_n). \quad (1)$$

These conditions determine every F_n . The F_n are the Fibonacci polynomials.

Lemma 1. $F_n(z) = \sum_{j \geq 0} \binom{n-1-j}{j} z^{n-1-2j}$ for $n \geq 1$. In particular, for $n = 2k + 1$,

$$\nabla_{T(2,2k+1)}(z) = \sum_{i=0}^k \binom{k+i}{2i} z^{2i}.$$

Proof. Write G_n for the right-hand side. Then $G_1 = 1$ and $G_2 = z$, and (1) gives $F_2 = F_0 + zF_1 = z$. For $n \geq 1$, the coefficient of z^{n+1-2j} in $G_n + zG_{n+1}$ is $\binom{n-j}{j-1} + \binom{n-j}{j} = \binom{n+1-j}{j}$ by Pascal's rule, which is the coefficient in G_{n+2} . So G_n satisfies the same recursion as F_n , and $F_n = G_n$ by induction. For $n = 2k + 1$, put $i = k - j$. The exponent is $2k - 2j = 2i$ and the coefficient is $\binom{2k-j}{j} = \binom{k+i}{k-i} = \binom{k+i}{2i}$. $\square$

As an independent check, the Alexander polynomial of a torus knot is $\Delta_{T(p,q)}(t) = \frac{(t^{pq}-1)(t-1)}{(t^p-1)(t^q-1)}$, and $\nabla(t^{1/2}-t^{-1/2}) = \Delta(t)$. The script `verify.py` derives the Conway polynomial from this formula and confirms that it agrees with (1) and Lemma 1 for all $T(2, n)$, $n \leq 61$.

n	coefficients of $z^0, z^2, z^4, \dots$ in $\nabla_{T(2,n)}$	peak index	$\lfloor (n-1)/4 \rfloor$
3	1, 1	0, 1	0
5	1, 3, 1	1	1
7	1, 6, 5, 1	1	1
9	1, 10, 15, 7, 1	2	2
13	1, 21, 70, 84, 45, 11, 1	3	3
25	1, 78, 1001, 5005, 12870, 19448 , 18564, 11628, $\dots$	5	6

Table 1: Even Conway coefficients of $T(2, n)$ and the predicted peak index.

3 The counterexamples

Theorem 2. For $T(2, 25)$ we have $(p-1)(q-1)/4 = 6$, and

$$\begin{aligned} \nabla_{T(2,25)}(z) = & 1 + 78z^2 + 1001z^4 + 5005z^6 + 12870z^8 + 19448z^{10} + 18564z^{12} \\ & + 11628z^{14} + 4845z^{16} + 1330z^{18} + 231z^{20} + 23z^{22} + z^{24}. \end{aligned}$$

The coefficient 19448 of z^{10} is strictly larger than the coefficient 18564 of z^{12} (position 6 under (R2)) and the coefficient 5005 of z^6 (position 6 under (R1)). So the peak-position clause fails for $T(2, 25)$ under readings (R1) and (R2).

Proof. By Lemma 1 with $k = 12$, the coefficient of z^{2i} is $\binom{12+i}{2i}$. This gives the list above. For example, $\binom{17}{10} = 19448$, $\binom{18}{12} = 18564$ and $\binom{15}{6} = 5005$. $\square$

For (R2), the ratio of consecutive coefficients is $\binom{k+i+1}{2i+2} / \binom{k+i}{2i} = \frac{(k+i+1)(k-i)}{(2i+1)(2i+2)}$. For $k = 12$ and $i = 5$ it equals $18 \cdot 7 / (11 \cdot 12) = 126/132 < 1$. So the sequence is already decreasing from index 5 to index 6. In general the ratio is about $(k^2 - i^2)/(4i^2)$, so the peak index of $\binom{k+i}{2i}$ is about $k/\sqrt{5} \approx 0.447k$. The predicted index is $\lfloor (n-1)/4 \rfloor = \lfloor k/2 \rfloor$, so the gap between them grows linearly in k . $T(2, 25)$ and $T(3, 13)$ (both of genus 12) are the smallest-genus torus knots that violate (R2), according to a survey of all $T(p, q)$ with $p < q \leq 30$ in `verify.py`.

Theorem 3. *For $T(2, 61)$ we have $\lfloor 60/4 \rfloor = 15$. The unique largest coefficient of $\nabla_{T(2, 61)}$ is $[z^{26}] = \binom{43}{26} = 421171648758$, whereas $[z^{15}] = 0$, $[z^{28}] = \binom{44}{28} = 416714805914$ and $[z^{30}] = \binom{45}{30} = 344867425584$. So the predicted position 15 is not a peak under any of (R1), (R2), (R3).*

Proof. Lemma 1 with $k = 30$ gives these values. The coefficient of z^{15} is zero because only even powers occur. The values are checked exactly in Lean and in Python. $\square$

Remark 4. Reading (R3) happens to hold for $T(2, 25)$ (position 6 is z^{10}). It already fails for $T(2, 5)$, where position 1 is z^0 with coefficient $1 < 3$. It also fails for $T(2, 61)$. The other clauses of the conjecture (nonnegativity, log-unimodality) are not needed: the conjecture is a conjunction, and one false clause refutes it.

Theorem 5. *Conjecture 00000000603 is false.*

4 Lean formalization

The directory `lean4/` contains a self-contained Lean 4.19.0 project that uses only the core library (no Mathlib). Its contents are:

- **ConwaySkein c**: a family of polynomials in z , given as coefficient functions $\mathbb{N} \rightarrow \mathbb{Z}$, that satisfies (1) (unlink, unknot, skein relation for σ_1^{n+2}).
- **skein_unique**: every such family agrees, coefficient by coefficient, with the reference lists `ref n` computed by the recursion. **ref_skein**: the reference family satisfies the hypotheses, so they are consistent and the result is not vacuous.
- **ref_25**: the full coefficient list of $\nabla_{T(2, 25)}$. **ref_61_peak**: in $\nabla_{T(2, 61)}$ the coefficient of z^{26} is strictly larger than every other coefficient. Both are proved by kernel `decide`.
- **PeakClaim c pos**: the conjecture's peak clause for all $T(2, n)$ (n odd, $n \geq 3$), with position $\lfloor (2-1)(n-1)/4 \rfloor$ and the indexing convention `pos` $\in \{(R1), (R2), (R3)\}$.
- **conjecture_00000000603_false**: for every family satisfying the skein facts, **PeakClaim** fails under all three readings. `T_2_25_counterexample` records the smallest example for (R1) and (R2).

`lake build` succeeds. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` reports at most `propext` and `Quot.sound`. The coefficient computations `ref_25` and `ref_61_peak` depend on no axioms at all.

5 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

References

- [1] J. H. Conway, An enumeration of knots and links, and some of their algebraic properties, in *Computational Problems in Abstract Algebra*, Pergamon, 1970, 329–358.
- [2] L. H. Kauffman, The Conway polynomial, *Topology* 20 (1981), 101–108.
- [3] D. Rolfsen, *Knots and Links*, Publish or Perish, 1976 (Alexander polynomial of torus knots).